# Application - Insight Development Grants

## Identification (mandatory)

### Applicant

**Family Name:** Minaker

**First Name:** Leia

**Middle Names:** M

**Current Position:** Director

**Primary Affiliation:** University of Waterloo

**Department/Division:** Future Cities Institute

### Application

**Application Title**

Enabling evidence-based policymaking and interdisciplinary social change research for homelessness: Developing and evaluating CivicInquire, Canada's first intersectional AI-powered social science platform

**Language of the Application**

(•) English ( ) French

**Committee**

23 - Multidisciplinary or interdisciplinary social sciences

**Joint or Special Initiative**

**Is this a research-creation project?**

( ) Yes (•) No

**Does your proposal involve Indigenous research as defined by SSHRC?**

( ) Yes (•) No

### Scholar type

| | |
|---|---|
| **Are you an Emerging Scholar or Established Scholar?** | Established |
| Confirmed Scholar type | Established |

## Established scholars: proposed versus ongoing research (mandatory)

I am an interdisciplinary social sciences researcher with expertise in disease prevention, epidemiology, health policy, public health, social determinants of health, and urban health. Using methods and theories from these fields, my research program has previously investigated food retail environments, food marketing, population dietary risk, public health surveillance for chronic diseas risk, youth health behaviours and beliefs, healthy city planning, smart cities and health inerventions, and equity in social policy and local and regional planning processes.

The proposed project builds on this work of my research program but, importantly, expands into several new areas of interdisciplinary social science research and knowledge dissemination activities. This can be illustrated in multiple aspects of the project. First, the propsoed project requires the development of a new research team that features expertise in areas that I do not have extensive previous collaboration with, notably bioinformatics, digital health, housing/homelessness, housing policy, and policy intensity analysis (esp. AI in policy reviews). Second, the proposed project applies an intersectional lens to AI methods in social science research and planning policy analysis, both of which reflect novel applications of intersectional theory. As a result, the proposed project will generate some of the first intersectional planning policy analyses quantifying meulti-level government alignment (O2) and critically investigating the state of housing and homelessness policy (O3) in Canada. And third, the proposed project aims to create an enduring open-access resource for any individual (e.g., academia, advocacy, housing developer, public sector) to use. This aim, however, will require ongoing maintenance and oversight. Such support will be provided by the Future Cities Institute, of which I am the director. The FCI is a new institute on the University of Waterloo and, as a part of its mission statement, is committed to producing public education resources, which this project will represent one of the first such resources created and maintained by the FCI. Therefore, this project is also intended to become a template for future public education resources that the FCI will develop in the coming years.

This project is in the early stages of the research process. To date, we have developed a prototype of the CivicInquire platform (www.civicinquire.com) via a small working group of collaborators involved this application. We have not collected, reviewed, or analyzed any data related to the objectives of the proposed project.

## Administering organization (individual grants) or lead organization (institutional grants)

| | |
|---|---|
| **Organization** | University of Waterloo |
| **Department/Division** | Planning |

## Invitations

| Role | Last Name | First Name | Organization | Department |
|---|---|---|---|---|
| Co-applicant | Butt | Zahid | University of Waterloo | School of Public Health Sciences |
| Co-applicant | Buttazzoni | Adrian | University of Waterloo | Planning |
| Co-applicant | Chen | Helen | University of Waterloo | School of Public Health Sciences |

| Role | Last Name | First Name | Organization | Department |
|---|---|---|---|---|
| Co-applicant | Da Silva E Souza Miranda | Pedro Augusto | University of Waterloo | Health |
| Co-applicant | Glover | Troy | University of Waterloo | Recreation and Leisure Studies |
| Co-applicant | Layton | Anita | University of Waterloo | Mathematics |
| Co-applicant | Mazumder | Robin | University of Waterloo | Planning |
| Collaborator | Gash | Leslie | No Primary Affiliation | No Department/Division |
| Collaborator | McEvoy | Chris | No Primary Affiliation | No Department/Division |

## Activity details

### Certification requirements (mandatory)

**Does the proposed research involve humans as research participants?** (●) Yes ( ) No

### Impact assessment (mandatory)

**Will any phase of the proposed research take place outdoors?** ( ) Yes (●) No

### Sensitive Technology Research Areas

**Will the proposed research and related activities funded by this grant aim to advance any of the listed STRAs? (required)** ( ) Yes (●) No

**Note:** If you answered "Yes" to the above question, you must complete and attach Attestation for Research Aiming to Advance Sensitive Technology Research Areas forms to your grant application — in the **STRAC Attestation Attachment section** — to comply with the policy.

If you answered “No”, you will not need to take any further steps under this policy.

### Keywords (mandatory)

Housing, Homelessness, Social Policy, Policy Analysis, Planning, Mixed Methods, Digital Infrastructure, AI, Intersectional Theory, Knowledge Mobilization

### Disciplines (mandatory)

1. Urban and Regional Studies, Environmental Studies Urban Planning
2. Natural Sciences and Engineering Computer and Information Science and Operations Research
3. Geography Urban Geography

## Areas of research (mandatory)

1. Housing
2. Social development and welfare
3. Science and technology

## Temporal periods

| | From | | To | |
|---|---|---|---|---|
| | Year | Period | Year | Period |
| 1. | 2026 | AD | 2028 | AD |
| 2. | | | | |

## Geographical regions

1. North America
2.
3.

## Countries

1. Canada
2.
3.
4.
5.

## Revisions since previous application

Over the past eight months our team has developed an initial version of the CivicInquire platform (www.civicinquire.uwaterloo.ca/search) and have been beta testing the performance and functionality of the platform and some of its core features (e.g., policy database search engine). However, work to-date has only resulted in our team creating a proof-of-concept for the CivicInquire platform. CivicInquire has not yet been comprehensively or formally reviewed for its comprehensiveness, communication clarity, information relevance, or user experience, nor has it been used for academic research. These activities are the primary focus and objectives of this proposed project.

## Summary of proposal

Homelessness, a downstream result of structural inequity across multiple systems including housing, healthcare, and planning, has major impacts on community cohesion, development, justice, and health, and is growing in Canada. Recent polling has reported that Canadians view the country's current direction regarding homelessness as being overwhelmingly negative, with >70% indicating homelessness is 'a growing issue.' In response, all levels of government have mobilized to combat homelessness; for instance, federal spending on homelessness programs has increased 374% over the past 10 years, while municipalities across the country have begun developing and implementing various types of homelessness prevention and affordable housing policies. Despite these efforts, Canadian cities continue to grapple with homeless encampments while trying to provide services for local unhoused populations. Given the extreme diversity of Canadian municipalities (e.g., large metropolitan cities, growing mid-sized municipalities, rural/northern/remote communities), the absence of shared tools and infrastructure to systematically examine, compare, and learn from homelessness policies across Canada's highly diverse municipal contexts represents a serious challenge that warrants attention.

The proposed multi-methods research uses machine learning (ML), artificial intelligence (AI), and qualitative key informant interviews to develop and evaluate a novel online, interactive platform for homelessness policy across Canada (i.e., CivicInquire). CivicInquire will leverage web-crawling and web-scraping capabilities to create a national homelessness policy database, a national policy dashboard, and multiple interactive visualization tools to present policy information for

academics, advocates, planners and health practitioners, municipal decision-makers, and the public. The proposed project has three research objectives. First (Objective 1; O1), we will evaluate and build upon a prototype of the CivicInquire platform (www.civicinquire.com) that our team has developed in recent months by conducting a global quality evaluation to assess its content, design, organization, and user-friendliness. Second (O2), we will construct a web scraping tool to identify and download all homelessness and housing policies and strategies from Canadian municipalities (N~1000) using the CivicInquire platform. Here, we will develop an ML algorithm designed to analyze the policy intensity (e.g., alignment in scope, strategies, goals/outcomes) of the collected data. Third (O3), we will conduct a critical discourse analysis of how key municipal stakeholders interpret the current state of homelessness and housing policy development, implementation, and impact in Canada with ~40 participants from diverse municipal contexts.

The CivcInquire platform will serve as a foundation for accelerating Canadian leadership in homelessness and housing policy research and action. Specifically, the open-access policy database, data visualization tools, and national dashboard will make CivicInquire the leading platform to support well-aligned, multisectoral, and effective policy approaches to the structural and systems-level prevention of homelessness in diverse Canadian communities. We anticipate that Canadian municipalities will use our platform to inform the development of their own policies, while researchers can use our platform and its open-access database to conduct future cutting-edge research exploring homelessness and/or housing policy impacts on multiple important outcomes. Our partners will help to facilitate our KMb plan in diverse municipal contexts including Toronto (large metro city), the Region of Waterloo (mid-sized, rural municipalities) and Nunavut (northern, remote communities).

## Roles and responsibilities

Our interdisciplinary team possess expertise and experience in homelessness policy and action, public health research, urban planning (O1-O3), machine learning and quantitative methods (O2), and qualitative and interview methods (O3), and health and social policy (O1-O3).

NPA Minaker (20% contribution) is an Associate Professor in the School of Planning (SoP) and is the Director of the Future Cities Institute (FCI) at the University of Waterloo (UW). As the nominated principal applicant (NPA), she will schedule and chair monthly meetings with the collective research team to discuss ongoing work and updates, as well as future directions and project milestones; oversee the team's progress toward the achievement of project objectives (O1-O3) with respect to agreed upon work timelines, deliverables, and trainee supervision; assist in the writing/reviewing of the manuscripts and dissemination of the findings (O2-O3); and advise on the recruitment, data collection, data analysis for O2-O3. She will also lead the supervision of all co-op students on the project.

Collaborator Gash (5% contribution; Toronto Shelter Network, TSN) is Executive Director of the TSN and a Registered Professional Planner with >30 years experience in stakeholder engagement and community development in the non-profit and public housing sectors. As a collaborator, Ms. Gash will contribute through participating in monthly team meetings, assisting the research team in capturing relevant criteria in the development of the policy intensity machine learning (ML) algorithm (O2), facilitating the recruitment of participants for the qualitative interviews through her existing work networks (O3), advising on interpretation of qualitative data (as appropriate) (O3), and support the knowledge mobilization (KMb) plan.

Collaborator McEvoy (5% contribution; Region of Waterloo, RoW) is the Manager of Housing Policy & Homelessness Prevention at the RoW and is responsible for implementing the Region's Plan to End Chronic Homelessness. He will participate in monthly team meetings to advise and guide the research team on refining research questions (O1), as well as elements of homelessness policy that are most relevant to capture from the perspective of mid-sized and rural municipalities (O2-O3). Additionally, he will support the recruitment of participants for the qualitative interviews (O3) through connecting the team with the Canadian Alliance to End Homelessness, and be involved in the KMb plans.

Co-A Butt (5% contribution) is an Assistant Professor at UW and a Canada Research Chair with expertise in digital and spatial epidemiology, homelessness, and health informatics. He will contribute to the project primarily though participating in monthly research team meetings, assisting in student supervision on O2, as well as advising and supporting the quantitative methods and analyses of O2 (e.g., development of the policy intensity ML algorithm (O2)) and O3 (e.g., interview topic). He will also advise on specifics of the tool development of O3 (i.e., interview guide questions, probes) and assist with manuscript writing/reviewing.

Co-A Buttazzoni (15% contribution) is an SSHRC PDF in the SoP at UW with expertise in qualitative methods, health policy, and urban planning. He will principally contribute to O1 and O3 via supporting the platform testing and evaluation analysis

(O1) and co-leading O3 and thus overseeing tool development (i.e., interview guide), recruitment, data collection and analysis, and manuscript writing. He will also assist with O2 through advising on policy intensity analysis matters related to data collection and aggregation (O2), as well as provide secondary supervision of the co-op students working on O1 and O3.

Co-A Chen (15% contribution) is an Associate Professor in Public Health Sciences at UW with expertise in AI for public and population health, AI learning for health systems, and information systems design and management. Her primary contributions will be delivered through serving as the co-lead researcher for O1 and O2, and a primary supervisor of all involved trainees. She will also lead the team in the development and implementation of the policy intensity ML algorithm, supervise and advise all team members involved in the data scraping and aggregating, the policy intensity analysis (O2), and writing/reviewing of academic articles (O2-O3). She will also help with the supervision of students working on O1 and O3.

Co-A Glover (8% contribution) is a Professor in the Department of Recreation and Leisure Studies at UW and the Director of the Healthy Communities Research Network. He has expertise concerning social infrastructure, placemaking, municipal policy analysis, the social impacts of urban planning, and advancing healthy communities through evidence-informed policy recommendations. His primary contributions will be attending monthly team meetings and supporting all research activities related to O3 (e.g., advising on data collection, participant recruitment, interpretation of findings), as well as helping to supervising the co-op student helping on O3.

Co-A Layton (7% contribution) is the Canada 150 Research Chair in Mathematical Biology and Medicine, and Professor of Applied Mathematics, Computer Science, Pharmacy and Biology at UW. She has expertise in the development and application of data-driven models to understand physiology and pathophysiology health outcomes. She will contribute through participating in monthly research team meetings, advising on the quantitative methods and analyses of O2, providing supervisory support for trainees working on O1 and O2, in addition to assisting in manuscript writing/reviewing. She will also help with supervision of students in O1 and O2.

Co-A Mazumder (5% contribution) is a PDF at the FCI at UW, and a former occupational therapist with direct experience working with vulnerable populations, including PEH. Leveraging this experience, as well as his expertise with ML applications for urban analytics, he will contribute through co-leading O2 (e.g., development of the policy intensity ML algorithm), supporting research activities related to O1 and O3 (e.g., interpretation of findings, manuscript writing/revising), and the supervision of co-op students working on O2.

Co-A Miranda (10% contribution) is a PhD candidate in the School of Public Health Sciences at UW with expertise in bioinformatics, digital health, and software engineering. He will be the lead designer and engineer that will co-lead all O1 activities (e.g., policy database search engine updated, embedding of interactive policy maps). He will also support all aspects of the data collection and analyses for O2 (e.g., framing/interpretation of findings). He will also help with academic article writing/reviewing and the supervision of co-op students working on O1 and O2.

## Roles and training of students

Pursuant to the methodological outline in the main application, one upper-year Math co-op student with experience in web development will assist with O1, two upper-year Math co-op students with experience in machine learning will assist with O2, and one upper-year co-op Planning student with expertise in qualitative methods will support all research activities related to O3. All hired trainees will serve as co-op students hired by the FCI, be directly supervised by FCI faculty associates (i.e., NPA, Co-As), and will be provided one-term (i.e., 4 month) co-op education opportunities that will be centred on gaining real-world work experiences related to urban planning, public health, data analytics for decision-making, policy analyses, and/or research methods. As such, our team will afford these students a number of opportunities to develop different applied research (e.g., qualitative data collection and analysis in O3), technical (e.g., digital platform development, evaluation in O1), knowledge mobilization (e.g., KMb tool/resource and digital infrastructure development in O1, partnership development across all stages) and communication (e.g., site/platform updates to the team during meetings) skills that they can use during their future academic endeavours.

More specifically, regarding O1, the co-op student will be tasked with undertaking several technical tasks that will intellectually challenge them with respect to evolving their initiative, time management, problem solving, and programming skills. Importantly, these two co-op students will be involved in developing and implementing programming strategies to support search engine optimization for the policy database (e.g., advanced search functions/MeSH term options), as well as adding suggested search filters (e.g., dropdown boxes for year-specific results, exclusion/inclusion features related to

source material) to the platform's database search; improving web crawling and web scraping ML tools that are central to the collection and aggregation of policy information from online sources, in addition to the completion of monthly updates to keep the policy database current; and creating and implemented requested resource-specific aides (e.g., FAQ tab, search result export functions) that seek to improve user experience. Such experiences should complement their academic training through providing multiple tasks that will develop their technical and problem-solving skills in direct relation to data and computational methods being leveraged for local decision-making processes.

Concerning O3, this co-op student will be foremost responsible for fulfilling duties related to academic research and knowledge mobilization (KMb) to varied target audiences (e.g., planners, local economic development officials, public health practitioners). Particular tasks will involve data collection via interviews and qualitative analysis via inductive thematic methodologies, advising on interview guide legibility/suitability, and developing article/study brief findings to disseminate to participants and collaborators. Subsequent KMb activities related to the objective will involve the student interacting with and presenting data visualizations and interactive tool (e.g., policy maps on the platform) overviews to interdisciplinary audiences of planners, public health practitioners, and municipal decision-makers.

This work will be intended to complement the co-op student's academic training through providing them 'real world' research experiences that will familiarize them with primary research process, the process and limitation of knowledge generation, and examples of applied (i.e., working directly with knowledge users) KMb strategies.

## Knowledge mobilization plan

CivicInquire serves as our central knowledge mobilization vehicle and features an open-access policy database, interactive maps, data visualizations, and a national dashboard. We commit to updating CivicInquire annually for at least five years post-grant, ensuring sustained impact as policies evolve. Primary audiences include municipal policymakers and planners using CivicInquire for evidence-based policy development; researchers accessing our database for interdisciplinary research; and non-governmental organizations advocating for homelessness prevention.

Our broader knowledge mobilization strategy emphasizes a bidirectional process of sustained engagement and iterative communication between our research team and knowledge users (KUs) and prioritizes co-creation throughout all phases to ensure research results are accessible, actionable and align Canada's homelessness and housing policy across diverse municipalities.

In O1, KUs (researchers, advocates, planners, and municipal representatives) will provide feedback on platform design. In O2, we will collaborate with partners from Toronto, Waterloo Region, and Nunavut to develop the ML algorithm driving policy intensity analyses. In O3, we will engage KUs through interviews exploring policy gaps and opportunities.

Traditional outputs include peer-reviewed manuscripts and academic conferences. Non-traditional dissemination targets decision-makers through presentations to the Federation of Canadian Municipalities and the Association of Municipalities of Ontario, with partners' networks amplifying reach across diverse contexts.

By grounding platform development in iterative co-design with KUs, CivicInquire will be more meaningful and responsive to real-world needs, ultimately supporting vulnerable Canadians through evidence-informed, multisectoral approaches to preventing chronic homelessness.

## Expected outcomes

### Scholarly benefits

1. Enhanced research collaboration
2. Enhanced research methods
3. Knowledge creation/intellectual outcomes

### Summary of expected scholarly outcomes

Our team will develop a sustainable AI-powered, data-aggregating, online homelessness policy platform (i.e., CivicInquire) with a national policy database and dashboard, which will be updated with new reviews, data visualization tools, and other knowledge mobilization (KMb) materials for researchers across the social sciences. The CivicInquire policy database will be updated on a monthly basis (via web-crawling, data-scraping tools) and maintained on an ongoing basis to ensure its accuracy, relevance, and timeliness for researchers and advocates in the knowledge-generating space. The analyses of O2 (policy intensity review) and O3 (policy discourse analysis) will offer researchers a novel quantification of homelessness policy alignment in reference to multiple intersectional considerations, and which will be relevant to communities across Canada (O2), and critical discussions of multisectoral contributions to current policy gaps and opportunities to remedy such shortfalls (O3).

## Societal benefits

1. Enhanced policy
2. New or enhanced collaborations, partnerships
3. Enriched public discourse

## Summary of expected societal outcomes

The developed CivicInquire platform is being intentionally designed as a free-to-use resource such that any individual or organization interested in or working on homelessness issues in Canada may use it. Working with the Future Cities Institute, who will maintain and update the platform, this project is intended to facilitate improved study of homelessness in Canada, as well as inform public discussion and knowledge on this issue. Additionally, dissemination about the CivicInquire platform and its features, tools, and KMb resources will be promoted by our collaborators through their professional networks, resulting in knowledge of this project among actors in large metropolitan, mid-size, small city, rural, and remote/northern communities. Conference presentations to non-academic (e.g., Federation of Canadian Municipalities) and academic (e.g., Association of Collegiate Schools of Planning) groups will encourage multisector teams to leverage the platform for their future work.

## Audiences

1. Canadian government
   Municipal government
2. Practitioner/professional/industrial associations
3. Academic sector/peers
4. Not-for-profit/community organizations
5. General public

## Summary of benefits to potential target audiences

CivicInquire will be a free-to-use platform available to users from all relevant fields across Canada. Multisector collaborations of policymakers, practitioners, and advocates can leverage CivicInquire to develop suitably aligned, comprehensive, and effective policy that address the multilevel governance barriers to the prevention of homelessness in diverse Canadian communities. Researchers may use CivicInquire to identify novel social change policy topics for study, or to build interdisciplinary research teams investigating relevant wellbeing and support issues, such as recurrent homelessness experience prevention or effective affordable housing solutions that stabilize PEH living circumstances. Collaborations between policymakers, practitioners, and researchers may also use the research (e.g., O3) and reviews (e.g., O2) to identify local topics in need of further study and/or to deliberate on with respect to addressing policy alignment (e.g., consistency in objectives).

## Funds requested from SSHRC

Year 1

| Personnel Costs | | | |
|---|---|---|---|
| **Student Salaries and Benefits/Stipends** | **Number** | **Amount** | **Justification** |
| **Undergraduate** | 3 | $61,050.00 | Includes 3 upper year math co-op students at $30/h x 35h/week x 17 weeks + 14% vacation pay = $20,350/student. CO-OP 1 (Sept-Dec 26) will support activities related to O1 (See detailed description, Section 2.1). CO-OP 2 (Jan-Apr 27) and CO-OP 3 (May-Aug 27) will support activities related to O2 (See detailed description, section 2.2). |
| **Masters** | | | |
| **Doctoral** | | | |
| **Subtotal** | | **$61,050.00** | |
| **Non-Student Salaries** | **Number** | **Amount** | **Justification** |
| **Postdoctoral** | | | |
| **Professional/Technical Services** | 1 | $15,000.00 | Includes the cost of a software developer (0.1 FTE) to oversee maintenance and continuous development of the data server and supporting infrastructure. |
| **Other** | | | |
| **Subtotal** | | **$15,000.00** | |
| **Travel and Subsistence Costs for Research** | **Number** | **Amount** | **Justification** |
| **Applicant/Team Members** | | | |
| **Student(s)** | | | |
| **Subtotal** | | **$0.00** | |
| **Travel and Subsistence Costs for Dissemination** | **Number** | **Amount** | **Justification** |
| **Applicant/Team Members** | | | |
| **Student(s)** | | | |
| **Subtotal** | | **$0.00** | |
| **Other Expenses** | | **Amount** | **Justification** |
| **Supplies** | | | |

| | | | |
|---|---|---|---|
| **Non-disposable equipment** | | | |
| | | | |
| | | | |
| | | | |
| **Subtotal** | | **$0.00** | |
| **Grand Total Year 1** | | **$76,050.00** | |

## Year 2

| **Personnel Costs** | | | |
|---|---|---|---|
| **Student Salaries and Benefits/Stipends** | **Number** | **Amount** | **Justification** |
| **Undergraduate** | 1 | $17,000.00 | Includes 1 upper year planning co-op students at $25/h x 35h/week x 17 weeks + 14% vacation pay = $17,000/student. CO-OP 4 (Sept-Dec 27) will support activities related to O3 (See detailed description, Section 2.3). |
| **Masters** | | | |
| **Doctoral** | | | |
| **Subtotal** | | **$17,000.00** | |
| **Non-Student Salaries** | **Number** | **Amount** | **Justification** |
| **Postdoctoral** | | | |
| **Professional/Technical Services** | | | |
| **Other** | | | |
| **Subtotal** | | **$0.00** | |
| **Travel and Subsistence Costs for Research** | **Number** | **Amount** | **Justification** |
| **Applicant/Team Members** | 2 | $5,000.00 | Money will be used to cover partial costs for 2 team members to attend either the Federation of Canadian Municipalities conference and/or the Association of Municipalities of Ontario conference. The remainder of costs will be covered by researchers on the team as per agreements and previous practice. |
| **Student(s)** | | | |
| **Subtotal** | | **$5,000.00** | |
| **Travel and Subsistence Costs for Dissemination** | **Number** | **Amount** | **Justification** |
| | | | |

| | | | |
|---|---|---|---|
| **Applicant/Team Members** | 1 | $1,950.00 | Money will be used to cover the fees associated with submitting/publishing scholarly manuscripts. The request for funds associated with Open Access Publication fees are minimal, given that researchers on our team will contribute additional funds as per our team's agreements and previous practice. |
| **Student(s)** | | | |
| **Subtotal** | | **$1,950.00** | |

| **Other Expenses** | **Amount** | **Justification** |
|---|---|---|
| **Supplies** | | |
| **Non-disposable equipment** | | |
| | | |
| | | |
| | | |
| **Subtotal** | **$0.00** | |
| **Grand Total Year 2** | **$23,950.00** | |
| **Grand Total** | **$100,000.00** | |

## Funds from other sources

| Funding Source | Contribution Type | Confirmed | Year 1 | Year 2 | Total |
|---|---|---|---|---|---|
| | | | | | $0.00 |
| Details | | | | | |
| | | | | | $0.00 |
| Details | | | | | |
| | | | | | $0.00 |
| Details | | | | | |
| Grand Total | | | | | $0.00 |

## Reviewer exclusion (if applicable)

### Excluded reviewers

| Exclusion Type | Family Name / Collaboration | First Name | Initials | Organization | Department | Email |
|---|---|---|---|---|---|---|
| No records to display. | | | | | | |

**1. CONTEXT & OBJECTIVES**:

Lifetime data indicate more than one in ten (11.2%) Canadians—or approximately 1.7 million people—have experienced some form of homelessness,[1] with annual estimates suggesting roughly 235,000 Canadians are affected each year .[2] These numbers are projected to grow[3] with profound consequences for individual health, public systems, and community wellbeing. As multiple structural systems (e.g., economic, health, political, planning) struggle to effectively respond,[4] Canadians view the country's current direction negatively, with >70% recognizing homelessness as 'a growing issue.'[5] People experiencing homelessness (PEH) face substantially elevated risks of depression, drug use,[6] external cause mortality, and infections (e.g., Hep C) from intravenous drug use,[7] and also report greater emergency department usage.[8] These risks generate significant social and economic costs for municipalities and healthcare systems alike. In Toronto alone, the healthcare costs of homelessness are an estimated $69.8-99.7 million annually.[9]

In response, all levels of government have mobilized to address homelessness. Federally, the National Housing Strategy Act was passed in 2019[10] with the aim to support and monitor the right to housing in Canada,[11] while Infrastructure Canada has spent an estimated $561 million annually on homelessness programs, an increase of 374% over the prior 10 years.[12] Provincially, Manitoba has developed a new initiative for vulnerable populations,[13] while New Brunswick has established a "Housing for All" strategy centered on supporting low-income populations and reducing housing waitlists.[14] Locally, communities of all sizes have implemented affordable and public housing strategies, as is the case in New Westminster,[15] Saskatoon,[16] and Vancouver.[17] Despite the proliferation of policy responses, Canada lacks a systematic way to examine, compare, and learn from homelessness policies across jurisdictions. Commonly cited issues include poor coordination,[18] a lack of long-term commitment from central institutions, inadequate targets,[19] and a dearth of resources/strategies designed to stabilize PEH and/or reduce their risk of recurrent experiences.[20] As homelessness is estimated to cost the national economy ~$7 billion annually,[21] and long-term analyses argue that the relative inaction over recent decades has cost Canada nearly $50 billion,[22] there is a pressing need to develop effective and coordinated homeless policy in Canada.

The multi-scale, -system, and -causal nature of homelessness in Canada presents a 'wicked problem'—a vast and complex issue lacking a clear solution[23]—for policymakers, practitioners, and researchers. Effectively addressing 'wicked problems' like homelessness (i.e., "the situation of an individual, family or community without stable, safe, permanent, appropriate housing, or the immediate prospect, means and ability of acquiring it"[24]) requires policy precision and responsiveness, as well as coordination across multiple fields and services.[25] In Canada, a central and underexamined challenge is the absence of tools and infrastructure to systematically understand the nation's homelessness policy landscape. <u>To support such policy precision and alignment, coordinated action, and a more comprehensive long-term vision for homelessness in Canada, this project aims to build and optimize Canada's first publicly accessible, interdisciplinary, and artificial intelligence- (AI) powered digital homelessness infrastructure (i.e., CivicInquire)</u>. Our project will develop this platform <u>to encourage evidence-based, aligned, and effective social policymaking, as well as interdisciplinary social science research on homelessness</u> through three objectives:

**<u>Objective 1</u>** (O1): develop and optimize an interactive digital platform (CivicInquire) that web-crawls, aggregates, and presents data from homelessness policies, strategies, and webpages. This objective will establish the foundational research infrastructure necessary to make homelessness policy visible, comparable, and analyzable across jurisdictions, which is a necessary to enable theory-informed governance research.

**<u>Objective 2</u>** (O2): develop a machine learning (ML) tool to analyze the policy intensity of the Canadian municipal policies and strategies related to homelessness/housing as collected in O1, thereby enabling comparative analysis across municipal types. This objective will translate policy intensity from an abstract concept into a measurable construct that allows social scientists to examine variation, alignment, and institutional commitment in homelessness governance.

**Objective 3** (O3): conduct a critical discourse analysis of stakeholder views on homelessness policy and its in/direct role(s) in creating, sustaining, and growing/lessening inequities to examine not only what homelessness policies do across Canada, but how they are understood, justified, and enacted by those responsible for them. This objective will situate the computational findings of O2 within lived policy interpretation to reveal how governance actors understand, justify, and enact homelessness policy in practice.

**2. METHODOLOGY & TIMELINES**:

**2.1 CivicInquire Development, Testing, and Optimization** *(O1: Sep/26-Dec/26)*: Having developed a prototype online policy platform (www.civicinquire.com), the aim of O1 will be to optimize (i.e., technical functionality, efficiency of the web-crawling, data aggregation, and data visualization tools) the platform. In partnership with the Future Cities Institute (FCI) at the University of Waterloo (UW), which will host, maintain, and annually update the platform for a period of at least 5 years, NPA Minaker, and Co-As Chen, Miranda, Layton, and Buttazzoni, will oversee this objective and supervise one upper-year Math co-op student with experience in web development and computer programming via a two-step process.

**CivicInquire Platform Performance Testing, Launch (Step 1)**: We will ensure functionality, usefulness, performance, responsiveness, and security of the CivicInquire prototype platform before officially making it publicly accessible. Prior to evaluation, CivicInquire will be implemented as a modular, service-oriented platform that separates data acquisition, processing, and user-facing services to support iterative improvement without disrupting live access. The core pipeline will include ethical, rate-limited acquisition of publicly available policy documents and related municipal artifacts; automated text normalization, deduplication, and metadata enrichment; storage in an open-format analytics layer following a "lakehouse" pattern to support both BI and research workloads[26]; and indexed services that power search, filtering, and analytic views for knowledge users. To strengthen transparency, reproducibility, and re-use, CivicInquire will implement machine-readable metadata and provenance tracking at the document and transformation level. Core metadata fields will include source jurisdiction, publication date, retrieval timestamp, canonical URL, document type, language, and a version identifier that captures updates to the source content. Provenance will be recorded using a standardized provenance model so that downstream analyses can be traced to specific inputs and processing steps.[27]

Our evaluation will use an established global quality evaluation framework for digital platforms to assess the content, design, organization and user-friendliness of our prototype.[28] Specifically, we will measure our prototype against the following criteria in each of these four domains: (a) content (i.e., characteristics of platform information; e.g., ensuring up-to-date information, and accuracy of sources, establishing the credibility of the platform[29]), (b) design (i.e., platform design and ability to attract and engage users; e.g., appropriateness of site for users, multimedia/varied data sources/data visualization[30]), (c) organization (i.e., platform grouping, categorization, or structure of information; e.g., clear indexing/easy navigation of information, functional links/tools[29]), and (d) user-friendliness (i.e., platform responsiveness, reliability; e.g., consistent usability, trusted security, cross-browser support[31]). Web-crawler development, web scraping tool (content), data aggregation, visualization (content, design, organization), and policy database search functionality and filtering (organization), and embedding the platform (i.e., our host website; design, organization, user-friendliness) will also be completed during step 1.

During optimization and evaluation, accessibility will be integrated into front-end design and testing to ensure the platform is usable by diverse stakeholders, including people with disabilities and varying levels of digital literacy. The interface will be evaluated against Web Content Accessibility Guidelines (WCAG) 2.2, including keyboard navigation, meaningful headings and labels, sufficient color contrast, and error prevention and recovery patterns.[32] Accessibility issues will be tracked as part of quality assurance so that improvements are measurable and sustained across releases. Additionally, all web collection activities will follow an ethics- and compliance-informed protocol tailored to social science research. The crawler will implement explicit rate limiting, exponential backoff, and operational safeguards to minimize load and reduce the

likelihood of access disruption. The protocol will include systematic documentation of source constraints (including terms of service and redistribution limitations where applicable), and internal checks to avoid collection of personal data unless explicitly required and ethically justified. This approach reflects evidence-informed guidance for legally and ethically responsible web scraping and the role that platform governance (including terms of service) plays in regulating data collection methods.[33,34]

**CivicInquire Platform Knowledge User Group Review (Step 2)**: Once the reviewed and updated prototype website has met the global quality evaluation and features a working database of policies and with associated knowledge mobilization (KMb) tools, our team and collaborators will share the platform via our networks (e.g., FCI community partners) to encourage researchers, advocates (N=50), planners, health practitioners, municipalities (N=>100) across Canada to use the platform and provide feedback. Access to the platform's policy database, infographics, data visuals (e.g., comparative radar charts), and mapping tools (e.g., choropleth maps showing spatial variation in policy intensity) will be provided to all user groups to solicit feedback that will be collected and analyzed alongside the global evaluation framework and used to further update the platform prior to the ensuing objectives (i.e., O2 and O3).

**2.2 CivicInquire Homelessness Policy Intensity Review** *(O2: Jan/27-Aug/27)*: To build the credibility of the platform as a source of information (from O1), we will complete a policy intensity analysis and publish results on CivicInquire. Quantitative policy analyses can allow municipalities to monitor policy effectiveness and improve systems.[35] Relative to manual methods, machine learning (ML)-assisted policy analysis have been found to be more efficient and objective.[36] O2 will develop a ML algorithm to conduct an evaluation of "policy intensity," which is a core determinant of policy effectiveness and can serve as a foundation for policy impact evaluation. Policy intensity indicates the extent of regulatory intervention in a specific policy area, assessing the extent to which policies are enforced (highly relevant to homelessness policies). High policy intensity is achieved when a policy is specialized and has detailed measures. Policy intensity is considered a *continuous* indicator, providing insights into policies and enabling policymakers to make necessary adjustments for more effective goal attainment, an important functionality to assess progress on addressing homelessness.[36] NPA Minaker and Co-As Chen, Miranda, Mazumder, Layton, and Butt will supervise two upper-year Math students with ML expertise during O2.

Policy intensity is a framework[35], that that weighs complex policy instruments according to criteria (e.g., measurable policy targets, budgets, clear objectives and timelines, policy monitoring[35–39]). We will work with partners from three diverse municipal contexts engaged in homelessness prevention and action to ensure we capture relevant criteria in the development of the policy intensity ML algorithm for municipalities across Canada: the Toronto Shelter Network (large metropolitan city) and the Region of Waterloo Community Services (mid-sized and rural municipalities). We are also working with the City of Iqaluit on a number of projects and look forward to engaging them in this process (remote, northern communities – partnership pending). Our team will apply an intersectional lens[40] to the ML algorithm to examine the extent to which and how policies address diverse components of homelessness policy (e.g., sex/gender). Responsible AI documentation will be produced for both specific data and large language policy models. The curated corpus will be accompanied by a structured dataset datasheet describing policy or strategy objective motivation, composition, collection processes, recommended uses, and limitations.[41] Each model will include a model card documenting, for each web-crawled policy or strategy, the stated objective(s), evaluation procedure(s), performance by relevant subgroups, and known risks or failure modes, supporting transparent communication to partners and decision-makers and reducing the risk of misuse in contexts for which the model is not well suited.[42]

**Policy Data Collection and Aggregation (Step 1)**: Using our prototype platform hosted on www.civicinquire.com and our developed web-scraping tool (run by beautifulsoup4 in Python), we will collect and aggregate publicly available homeless/housing policy and strategy data (e.g., from municipal websites, legal databases, regulatory repositories [e.g., by-laws, housing

strategies related to homelessness]) from 2000-2026 from municipalities (N=3,573) and provincial/territorial governments (N=13) across Canada. Based on our pilot work, we estimate retrieving at least 1000 relevant municipal documents. Policies will subsequently be categorized by title (*specialized* [e.g., "Homelessness Reduction Plan"] or *comprehensive* [e.g., "Urban Development Strategy"]), by document type (based on legal status - legislation, municipal regulations) and by theme (e.g., emergency shelters, transitional housing, services).

**Policy Intensity Structure and Machine Learning Integration (Step 2)**: The policy intensity dimensions will be co-created with our partners who have extensive knowledge and experience related to homelessness in Canada. Intensity dimensions (e.g., measurable targets) will first be selected. Our team will then create scoring metrics specifically for the: i) policy title (using a regular expression algorithm, keywords related to homelessness and affordable housing will be identified; titles with certain terms will receive higher intensity scores), ii) document type (scores based on the types of the document [e.g., laws, guidelines]; higher scores assigned to legally binding documents), iii) policy measures (ML tools will evaluate how detailed measures are within a policy; documents with precise plans, timelines, and specific goals will receive higher scores), iv) implementation (scores based on the extent to which a document details a implementation plan; more detailed plans receive higher scores), and v) level of alignment (municipal policies will be assessed for relevant provincial/territorial and federal homelessness/housing strategy/policy links; more aligned policies receive higher scores). Next, we will further integrate ML into the policy intensity process to assist with text processing, scoring policy measures, and classifiers. During text processing, policy texts are cleaned by removing irrelevant content; in this step, we will create an ML that will develop a homelessness-specific lexicon (e.g., "supportive housing"). Regarding policy measures, documents will be scored using Latent Dirichlet Allocation (LDA)[43] to extract key topics (e.g., housing, health, shelters) and evaluate the specificity of measures. The LDA will provide a transparent baseline for discovering recurring policy themes and their evolution over time.[44] Where topic coherence or interpretability is limited by short texts or technical language, transformer-based sentence embeddings will be evaluated to support clustering and semantic retrieval that better reflects policy meaning rather than surface word overlap.[45]

Finally, to train the model, we will use Extreme Gradient Boosting (XGBoost[46]), a powerful classifier that combines several basic classifiers with varying weights into a strong classifier, and outperforms other classifiers in policy intensity research.[36] We will use common metrics, including accuracy, precision, recall, and F1-score (the harmonic average of recall and precision) to evaluate the performance[36] of our classifier models. Interpretability will be integrated to support trust, policy relevance, and error diagnosis. Feature attribution methods based on Shapley values will be used to explain which inputs drive classification decisions, to identify spurious signals (e.g., jurisdiction-specific boilerplate), and to inform targeted improvements to both the codebook and the feature engineering strategy.[47]

**Policy Intensity Calculation (Step 3)**: Last, we will calculate policy intensity by combining the scores from the three dimensions, where: $\text{intensity}_t = \sum(\text{type}_j \times \text{title}_j \times \text{measure}_j), j=1$. As policy intensity is associated with policy impact in prior research,[35–38] O2 activities will result in a large and state-of-the-art dataset of policy intensity scores for homelessness policies in Canada.

**2.3 Homelessness Policy Discourse Analysis** *(O3: Sep/27-Jun/28)*: To encourage interdisciplinary research, create novel and relevant information for practitioners and policy-makers, an upper-year co-op Planning student with expertise in discourse analysis will work under the supervision of NPA Minaker, and Co-As Glover and Buttazzoni. The aim of O3 will be to examine how stakeholders discursively construct homelessness as a policy problem, assign responsibility across sectors and scales, and define what constitutes 'effective' or 'aligned' policy action. Here, O3 does not simply contextualize policy intensity scores; it reveals how homelessness policy is interpreted, justified, and contested by actors operating within different institutional and professional logics.

**Recruitment and Eligibility (Step 1)**: Purposive sampling methods will be used to recruit an information-rich[48] multisectoral sample of stakeholders (n=~40) representing five sectors:

municipal decision makers, planning or public health practitioners, advocates, scholars, and developers. Eligible participants will be currently involved in addressing homelessness and/or providing affordable housing in Canada. To discuss homelessness in rural/remote, mid-sized, and metropolitan contexts, we will stratify recruitment through collaborator networks.

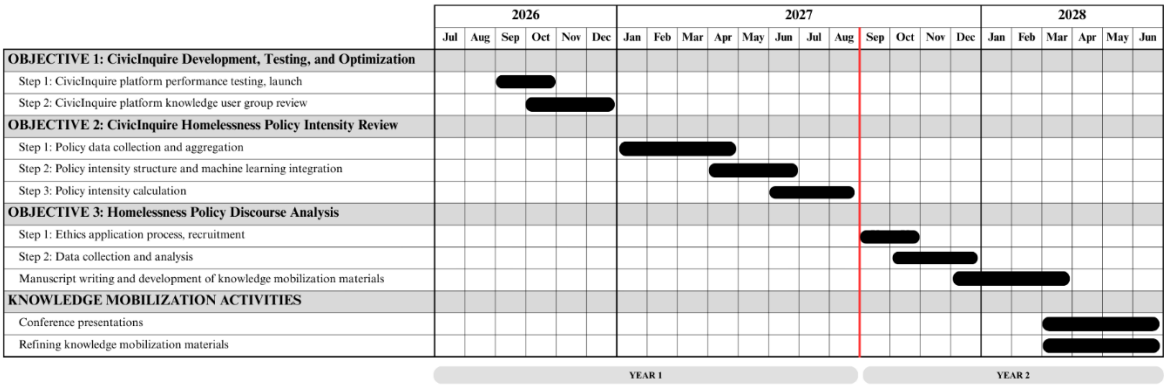
**Data Collection (Step 2)**: We will conduct ~40 (~8 participants per sector: municipal decision makers, planning and public health practitioners, advocates, scholars, and developers) 30–60-minute semi-structured virtual interviews with key stakeholders from across Canada. Semi-structured interviews allow for comparability across conversations (e.g., consistent framing questions) and capturing specific insights in response to participant-specific probes.[49] Our interview guide will be informed by the results of O2. For example, framing questions will inquire about the types of homelessness policies present in the respective communities of, or the current state of sectoral knowledge germane to, participants; the extent to which local policies align with provincial, territorial, federal policies and/or strategies that participants are familiar with; and the participants' perspectives of the different role(s) and responsibilities of diverse sectors in contributing to effective, evidence-based, and aligned homelessness policies.

**Data Analysis (Step 3)**: After the interviews, our analysis will be guided by intersectional theory[40] to conduct a critical discourse analysis (i.e., analyzing role(s) of discourse in the (re)production and challenge of social power via institutions/elites/groups that result in disparities).[50] Our discourse analysis will aim to systematically and comparatively investigate[51] the distinct ways stakeholders from multiple sectors: i) understand homelessness in their jurisdiction and sector, and the primary contributing policy factors that have created, maintained, and/or exacerbated or reduced the conditions (e.g., social, political, economic) contributing to priority contemporary homelessness/housing issues; ii) express shared assumptions (implicit, explicit) about the roles and responsibilities of their sector's professional beliefs and practices with respect to current policy gaps/limitations (esp. intersectional topics e.g., mental health); and iii) propose actions and initiatives to promote aligned effective policy to end homelessness. Findings will be compared to the results of O2 (quantitative policy intensity analysis) to examine how current municipal, provincial, and federal homelessness policy in Canada is perceived by key stakeholders across housing/homelessness fields to create structures and systems that affect action[51] on ending chronic homelessness across Canadian communities. Comparative analysis will examine points of convergence and divergence between measured policy intensity and stakeholder interpretations of policy alignment to reveal where technical robustness and perceived effectiveness diverge.

**3 Social Science Impacts**: CivicInquire will be a free-to-use platform available to users from all relevant fields across Canada. The Future Cities Institute (FCI) at the University of Waterloo is committed to maintaining and updating the platform and its developed resources on an ongoing basis. The FCI will promote the use of the CivicInquire platform among homelessness advocates, practitioners, policymakers, and researchers to improve the study of homelessness in Canadian communities, encourage the formation of multisectoral teams addressing local practice and policy deficiencies, and enhance public discussion and knowledge pertaining to this issue. Objective 1 will take several steps to establish the foundational research, data visualization, and KMb infrastructure required to make homelessness policy accessible, searchable, and usable to multisector audiences and improve homelessness governance practice and research. The policy intensity review findings of O2 will provide a novel quantification of homelessness policy alignment in reference to multiple intersectional considerations across Canadian cities of varied sizes. Objective 2 will therefore produce a novel example of translating policy intensity from a theoretical frame into a measurable construct for homelessness policy in Canada, consequently providing social scientists new information to analyze policy variation, alignment, and institutional commitment pertaining to homelessness. Lastly, the critical policy discourse findings of O3 will situate the quantitative policy intensity findings of O2 within the lived policy interpretations of stakeholders with academic, advocacy, municipal, and practitioner backgrounds to illuminate the different ways in which government, development, and political actors understand, justify, and enact homelessness policy in their professional fields.

**Figure 1. Project TIMELINES by Objective (July 2026 to June 2028)**

The primary activities of this project will proceed as follows:




**Objective 1: CivicInquire Development, Testing, and Optimization (Sept-Dec 2026)**

- Step 1: CivicInquire Platform Performance Testing, Launch (Sept-Oct 2026)
- Step 2: CivicInquire Platform Knowledge User Group Review (Oct-Dec 2026)

**Objective 2: CivicInquire Homelessness Policy Intensity Review (Jan-Aug 2027)**

- Step 1: Policy Data Collection and Aggregation (Jan-Apr 2027)
- Step 2: Policy Intensity Structure and Machine Learning Integration (Apr-Jun 2027)
- Step 3: Policy Intensity Calculation (Jun-Aug 2027)

**Objective 3: Homelessness Policy Discourse Analysis (Sep 2027 – Mar 2028)**

- Step 1: Ethics Application Process, Recruitment (Sep-Oct 2027)
- Step 2: Data Collection and Analysis (Oct-Dec 2027)
- Manuscript Writing and Development of Knowledge Mobilization Materials (Dec 2027 – Mar 2028)

**Knowledge Mobilization Activities**

- Conference presentations (Mar-Jun 2028)
- Refining Knowledge Mobilization Materials (Mar-Jun 2028)

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